

SHETH L.U.J AND SIR MV COLLEGE

PRACTICAL NO:-07

AIM : Sampling and Sampling Distributions

- Conduct a survey and collect data from a sample population, ensuring proper sampling techniques are employed.
- Use the Central Limit Theorem to analyze the sampling distribution of a sample mean and estimate population parameters.

```
T128_P_07.py - C:/Users/Rani/Downloads/T128_P_07.py (3.13.7)
File Edit Format Run Options Window Help

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Student Details
print("Name: Rani Chavre")
print("Roll No.: T128")
print("AIM: Sampling and Sampling Distributions")

# Load Dataset
df = pd.read_csv("Supermart Grocery Sales - Retail Analytics Dataset.csv")

# Population
population = df[["Profit"]].dropna()

print("\n--- Population Information ---")
print("Population Size:", len(population))
print("Population Mean:", round(population.mean(), 2))

# Simple Random Sampling
sample_size = 30
sample_means = []

for i in range(1000):
    sample = population.sample(n=sample_size)
    sample_means.append(sample.mean())

IDLE Shell 3.13.7*
Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (A
64)] on win32
Enter "help" below or click "Help" above for more information.

>>>

===== RESTART: C:/Users/Rani/Downloads/T128_P_07.py ==
=====
Name: Rani Chavre
Roll No.: T128
AIM: Sampling and Sampling Distributions

--- Population Information ---
Population Size: 9994
Population Mean: 374.94

--- Sampling Distribution ---
Sample Size: 30
Number of Samples: 1000
Sampling Mean: 374.89
Standard Error: 44.57
```

